

# Mohammad Muzakkir Jamal

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#)

## Career Objective

Computer Science undergraduate passionate about building reliable, scalable, and cloud-native software solutions. Experienced in backend development using Go, Kubernetes, Docker, PostgreSQL, and REST APIs, along with hands-on work in predictive autoscaling systems and AI-powered applications. Strong foundation in data structures, algorithms, and software engineering principles, complemented by practical experience in developing containerized applications and cloud-native architectures. Seeking an opportunity to contribute to challenging engineering projects while continuously enhancing my expertise in backend systems, cloud technologies, and distributed computing.

## Technical Skills

**Languages:** Go, C++, C, Python, JavaScript, HTML, CSS

**Cloud & DevOps:** AWS, Kubernetes, Docker, Minikube, CI/CD Fundamentals

**Backend & APIs:** Gin, Gorilla Mux, GORM, REST APIs, JWT Authentication

**Databases:** PostgreSQL, MongoDB, MySQL

**AI & Data:** LangChain, OpenAI API, Time-Series Forecasting

**Tools:** Git, GitHub, Postman, VS Code

**Areas of Interest:** Cloud Computing, DevOps, Cloud-Native Systems, Kubernetes, Backend Engineering

## Projects

**Predictive Autoscaler for Kubernetes**— Go, Kubernetes, Docker, React, Time-Series Forecasting

- Engineered a Kubernetes-native predictive autoscaling system that forecasts workload demand and generates proactive scaling recommendations.
- Developed a Go-based predictive scaler integrating real-time metric collection, adaptive forecasting, and Kubernetes-based scaling decisions, evaluated using MAE, RMSE, and MAPE under synthetic burst workloads.
- Integrated with the Kubernetes API using RBAC-secured access to manage application replica scaling.
- Containerized application components with Docker and deployed the complete system locally using Minikube.
- Developed a React dashboard to visualize actual versus predicted workloads, scaling recommendations, forecasting errors, and scaling events.
- Evaluated forecasting accuracy using MAE, RMSE, and MAPE under synthetic burst-traffic workloads.

**MagicStreamMovies**—Go, React, MongoDB, LangChain, OpenAI API

- Built a full-stack movie streaming platform with a scalable Go backend, MongoDB database, and React frontend.
- Developed RESTful APIs for movie management, authentication, and AI-powered recommendations using Go, MongoDB, JWT, and LangChain.
- Integrated an AI-driven recommendation system using LangChain and the OpenAI API to provide personalized movie suggestions.

## Education

B.Tech in Computer Science & Engineering — Jamia Hamdard (Expected 2027) — current CGPA — 8.13

Senior Secondary (12th) — Jamia Senior Secondary School, Jamia Millia Islamia — 76%

Higher Secondary (10th) — Jamia Senior Secondary School, Jamia Millia Islamia — 91.2%

## Achievements & Highlights

- Solved 250+ DSA problems across LeetCode and other coding platforms.
- Built and deployed cloud-native applications using Docker and Kubernetes.
- Authoring a research paper on predictive autoscaling in Kubernetes-based cloud-native systems; manuscript currently under review.
- Strong understanding of data structures, algorithms, object-oriented programming, and backend system design.